

Free-LCS V^* Variants and the Weak Sequential-Precompact Split

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Status and Source Problem

This packet gives a consolidated partial answer to Problem 3.18 of Saak Gabrielyan's paper *Gelfand–Phillips type properties of locally convex spaces* [1]. The problem asks:

and hence relatively compact. Thus $L(X)$ trivially has the coarse GP_p property. $\square$

Theorems 3.16 and 3.17 suggest the following problem.

Problem 3.18. *Let $1 \leq p \leq q \leq \infty$. Characterize Tychonoff spaces X for which $L(X)$ has one of the $V_{(p,q)}^*$ type properties defined in Definition 2.5.*

We know from (ii) of Lemma 3.1 that the case $(1, \infty)$ is the strongest one in the sense that if E

The first result below is a full characterization for the properties

$$V_{(p,q)}^*, \quad EV_{(p,q)}^*, \quad sV_{(p,q)}^*, \quad sEV_{(p,q)}^*$$

of the free locally convex space $L(X)$. The second result gives a positive criterion for the weak sequential-precompact variants $wsV_{(p,q)}^*$ and $wsEV_{(p,q)}^*$, and shows by the example $X = [0, 1]$ that these weak variants cannot have the same finite-functionally-bounded characterization as the stronger variants.

Definitions and Standard Facts Used

For a Tychonoff space X , write elements of the free locally convex space $L(X)$ as finite sums

$$\chi = a_1x_1 + \cdots + a_nx_n,$$

where $x_1, \dots, x_n \in X$ are distinct and $a_i \neq 0$. Put

$$\|\chi\|_1 = |a_1| + \cdots + |a_n|, \quad \text{supp}(\chi) = \{x_1, \dots, x_n\}.$$

For $A \subseteq L(X)$, let

$$\text{supp}(A) = \bigcup_{\chi \in A} \text{supp}(\chi), \quad C_A = \sup\{\|\chi\|_1 : \chi \in A\}.$$

The dual of $L(X)$ is $C(X)$, with pairing

$$\langle f, \chi \rangle = a_1f(x_1) + \cdots + a_nf(x_n).$$

We use the bounded-set criterion for free locally convex spaces recalled in Theorem 3.17 of [1] from Proposition 2.7 of [4]: a subset $B \subseteq L(X)$ is bounded if and only if $\text{supp}(B)$ is functionally bounded in X and $C_B < \infty$.

A nonempty set $A \subseteq E$ is a (p, q) - V^* set if, for every weakly p -summable sequence (χ_n) in the strong dual E'_β , the scalar sequence

$$\left(\sup_{a \in A} |\langle \chi_n, a \rangle| \right)_n$$

belongs to ℓ_q , if $q < \infty$, and to c_0 , if $q = \infty$. The (p, q) - EV^* sets are defined similarly, but only equicontinuous weakly p -summable sequences in E'_β are tested. We use Lemma 7.2 of [2]: every (p, q) - V^* set is a (p, q) - EV^* set, and every (p, q) - EV^* set is bounded.

Proof Intuition

The sufficient direction is finite-dimensional: if all functionally bounded subsets of X are finite, then every bounded subset of $L(X)$ has finite support and bounded coefficient norm. Since every EV^* -set is bounded, every V^* - or EV^* -set is relatively compact in a finite-dimensional coordinate span.

For the converse, an infinite functionally bounded set $D = \{x_i\}$ supports a sequence of finite measures

$$a_m = \sum_{i=0}^m 2^{-i-1} x_i \in L(X).$$

These partial sums behave like a norm-compact set after restriction to D , so they are (p, q) - V^* sets for every $p \leq q$. But in the weak topology of $L(X)$ they converge pointwise on $C(X)$ to the infinite measure $\sum_i 2^{-i-1} \delta_{x_i}$, which is continuous on the strong dual but is not an element of $L(X)$. Thus the sequence is a V^* witness that compactness and sequential compactness fail.

The weak sequential-precompact variants require a different idea. If a bounded sequence in $L(X)$ has countable support S , then it is a uniformly bounded sequence of functionals on the Banach space obtained by closing the restrictions $\{f|_S : f \in C(X)\}$ in $\ell_\infty(S)$. When that restriction space is separable, its dual ball is weak-star compact metrizable, and hence the original sequence has a weakly Cauchy subsequence.

Lemma 1 (Weighted partial sums are universal V^* sets). *Let $D = \{x_i : i \in \mathbb{N}\}$ be an infinite functionally bounded subset of a Tychonoff space X , and put $w_i = 2^{-i-1}$. For*

$$a_m = \sum_{i=0}^m w_i x_i \in L(X), \quad m \in \mathbb{N},$$

the set $A = \{a_m : m \in \mathbb{N}\}$ is a (p, q) - V^ subset of $L(X)$ for every $1 \leq p \leq q \leq \infty$.*

Proof. Since D is functionally bounded, the set $\{x_i : i \in \mathbb{N}\}$ is a bounded subset of $L(X)$. Hence the restriction map

$$R : L(X)'_\beta = C(X)_\beta \longrightarrow \ell_\infty, \quad R(f) = (f(x_i))_{i \in \mathbb{N}},$$

is continuous: its norm is the strong-dual seminorm of uniform convergence on the bounded set $\{x_i : i \in \mathbb{N}\}$.

Define a bounded operator $H : \ell_\infty \rightarrow \ell_\infty$ by

$$H(y)(m) = \sum_{i=0}^m w_i y(i), \quad y = (y(i))_{i \in \mathbb{N}}.$$

We first record that H sends weakly p -summable sequences in ℓ_∞ to norm ℓ_p -summable sequences when $p < \infty$, and sends weakly null sequences to norm-null sequences when $p = \infty$.

Let $1 \leq p < \infty$, and let (y_k) be weakly p -summable in ℓ_∞ . The map

$$S : (\ell_\infty)' \rightarrow \ell_p, \quad S(\varphi) = (\varphi(y_k))_{k \in \mathbb{N}},$$

is well-defined. Its graph is closed, so by the closed graph theorem S is bounded. If e'_i denotes the i -th coordinate functional on ℓ_∞ , then

$$\|(y_k(i))_{k \in \mathbb{N}}\|_{\ell_p} = \|S(e'_i)\|_{\ell_p} \leq \|S\| \quad \text{for every } i.$$

Minkowski's inequality gives

$$\begin{aligned} \left(\sum_k \|H(y_k)\|_\infty^p \right)^{1/p} &\leq \left(\sum_k \left(\sum_i w_i |y_k(i)| \right)^p \right)^{1/p} \\ &\leq \sum_i w_i \left(\sum_k |y_k(i)|^p \right)^{1/p} \leq \|S\| \sum_i w_i < \infty. \end{aligned}$$

Thus $(\|H(y_k)\|_\infty)_k \in \ell_p$, hence it belongs to ℓ_q for every $q \geq p$.

For $p = \infty$, the finite-rank truncations

$$H_N(y)(m) = \sum_{i=0}^{\min\{m, N\}} w_i y(i)$$

converge to H in operator norm, since $\|H - H_N\| \leq \sum_{i > N} w_i$. Therefore H is compact. Compact operators send weakly null sequences in a Banach space to norm-null sequences, so $\|H(y_k)\|_\infty \rightarrow 0$.

Now let (f_k) be weakly p -summable in $L(X)_\beta'$. By the continuity of R , the sequence $y_k = R(f_k)$ is weakly p -summable in ℓ_∞ . Moreover,

$$\sup_{a \in A} |\langle f_k, a \rangle| = \sup_m \left| \sum_{i=0}^m w_i f_k(x_i) \right| = \|H(y_k)\|_\infty.$$

The preceding paragraph proves that these suprema lie in ℓ_q , or in c_0 when $q = \infty$. Hence A is a (p, q) - V^* set. $\square$

Lemma 2 (The weighted partial sums are not weakly compact). *With D , w_i , a_m , and A as in Lemma 1, the set A is neither relatively weakly compact nor relatively weakly sequentially compact in $L(X)$.*

Proof. Define

$$\mu(f) = \sum_{i=0}^{\infty} w_i f(x_i), \quad f \in C(X).$$

The series is absolutely convergent because D is functionally bounded. Also $\mu \in L(X)'' = (L(X)'_\beta)'$: indeed, the seminorm $f \mapsto \sup_i |f(x_i)|$ is continuous on $L(X)'_\beta$, since $\{x_i : i \in \mathbb{N}\}$ is bounded in $L(X)$, and

$$|\mu(f)| \leq \left(\sum_i w_i \right) \sup_i |f(x_i)|.$$

For every $f \in C(X)$, one has

$$\langle f, a_m \rangle = \sum_{i=0}^m w_i f(x_i) \longrightarrow \mu(f).$$

The functional μ is not represented by any element of $L(X)$. If $\mu = \sum_{j=1}^r c_j y_j$ for finitely many $y_j \in X$, choose N with $x_N \notin \{y_1, \dots, y_r\}$. Since X is Tychonoff and the set $\{y_1, \dots, y_r, x_0, \dots, x_{N-1}\}$ is finite and closed, there is $f \in C(X, [0, 1])$ such that $f(x_N) = 1$ and f vanishes on that finite set. Then the finite sum gives $\sum_j c_j f(y_j) = 0$, while

$$\mu(f) = \sum_i w_i f(x_i) \geq w_N > 0,$$

a contradiction.

If A were relatively weakly compact, its weak closure K in $L(X)$ would be compact. The canonical evaluation map

$$J : L(X)_w \rightarrow \mathbb{F}^{C(X)}, \quad J(\chi) = (\langle f, \chi \rangle)_{f \in C(X)}$$

is a topological embedding. The compact set $J(K)$ is closed in the Hausdorff product $\mathbb{F}^{C(X)}$, contains $J(A)$, and therefore contains the product limit $(\mu(f))_{f \in C(X)}$ of the sequence $J(a_m)$. This would represent μ by an element of $L(X)$, contradicting the previous paragraph. Hence A is not relatively weakly compact.

The same pointwise-limit argument shows that no subsequence of (a_m) weakly converges in $L(X)$: any subsequential weak limit would again represent μ . Therefore A is not relatively weakly sequentially compact. $\square$

Theorem 1 (Characterization for V^* , EV^* , and their sequential variants). *Let $1 \leq p \leq q \leq \infty$, and let X be a Tychonoff space. The following assertions are equivalent:*

1. *every functionally bounded subset of X is finite;*
2. *$L(X)$ has the property $V_{(p,q)}^*$;*
3. *$L(X)$ has the property $EV_{(p,q)}^*$;*
4. *$L(X)$ has the property $sV_{(p,q)}^*$;*
5. *$L(X)$ has the property $sEV_{(p,q)}^*$.*

Moreover, if these conditions hold, then $L(X)$ also has $wsV_{(p,q)}^$ and $wsEV_{(p,q)}^*$.*

Proof. Assume first that every functionally bounded subset of X is finite. Let $B \subseteq L(X)$ be a (p, q) - EV^* set. By Lemma 7.2 of [2], B is bounded. The bounded-set criterion for $L(X)$ gives

$C_B < \infty$ and says that $\text{supp}(B)$ is functionally bounded; by the present hypothesis, $F = \text{supp}(B)$ is finite. Hence

$$B \subseteq \left\{ \sum_{x \in F} a_x x : \sum_{x \in F} |a_x| \leq C_B \right\}.$$

The set on the right is compact in the finite-dimensional space $\text{span}(F)$. Thus every (p, q) - EV^* set is relatively compact, hence relatively weakly compact, relatively weakly sequentially compact, and weakly sequentially precompact. Since every (p, q) - V^* set is a (p, q) - EV^* set, all the listed V^* , EV^* , sequential, and weak sequential-precompact properties follow.

Conversely, suppose that X contains an infinite functionally bounded subset. Lemmas 1 and 2 give a (p, q) - V^* set $A \subseteq L(X)$ which is neither relatively weakly compact nor relatively weakly sequentially compact. Since every (p, q) - V^* set is a (p, q) - EV^* set, the same A also witnesses the failure of $EV^*_{(p,q)}$ and $sEV^*_{(p,q)}$. Thus none of assertions (2)–(5) can hold. $\square$

A Positive Criterion for the Weak Variants

The preceding theorem shows that finite functional boundedness is sufficient for $wsV^*_{(p,q)}$ and $wsEV^*_{(p,q)}$. It is not necessary. We record a broader sufficient condition which covers all compact metrizable spaces and gives a concrete separation example.

Theorem 2 (Restriction separability gives weak sequential precompactness). *Let X be a Tychonoff space. Assume that for every countable functionally bounded set $S \subseteq X$, the Banach space*

$$E_S = \overline{\{f|_S : f \in C(X)\}}^{\|\cdot\|_\infty} \subseteq \ell_\infty(S)$$

is separable. Then every bounded subset of $L(X)$ is weakly sequentially precompact.

Proof. Let $B \subseteq L(X)$ be bounded, and let (χ_n) be a sequence in B . By the bounded-set criterion, $C_B < \infty$ and $\text{supp}(B)$ is functionally bounded. Put

$$S = \bigcup_{n \in \mathbb{N}} \text{supp}(\chi_n).$$

Then S is countable and functionally bounded. By hypothesis,

$$E_S = \overline{\{f|_S : f \in C(X)\}}^{\|\cdot\|_\infty}$$

is a separable Banach space.

Write

$$\chi_n = \sum_{x \in F_n} a_{n,x} x, \quad F_n \subseteq S \text{ finite.}$$

Each χ_n defines a continuous linear functional $\mu_n \in E'_S$ by

$$\mu_n(g) = \sum_{x \in F_n} a_{n,x} g(x), \quad g \in E_S.$$

Moreover

$$\|\mu_n\| \leq \sum_{x \in F_n} |a_{n,x}| = \|\chi_n\|_1 \leq C_B.$$

Thus (μ_n) is contained in the weak-star compact set $C_B B_{E'_S}$. Since E_S is separable, this compact set is metrizable for the weak-star topology. Hence (μ_n) has a weak-star convergent subsequence, say (μ_{n_k}) .

For every $f \in C(X)$, its restriction $f|_S$ belongs to E_S , so the scalar sequence

$$\langle f, \chi_{n_k} \rangle = \sum_{x \in F_{n_k}} a_{n_k, x} f(x) = \mu_{n_k}(f|_S)$$

converges. Hence (χ_{n_k}) is weakly Cauchy in $L(X)$. Since the original sequence in B was arbitrary, B is weakly sequentially precompact. $\square$

Corollary 1 (Weak V^* variants under restriction separability). *Under the hypothesis of Theorem 2, $L(X)$ has $wsV^*_{(p,q)}$ and $wsEV^*_{(p,q)}$ for every $1 \leq p \leq q \leq \infty$.*

Proof. Every (p, q) - V^* set is a (p, q) - EV^* set, and every (p, q) - EV^* set is bounded. By Theorem 2, every bounded subset of $L(X)$ is weakly sequentially precompact. $\square$

Corollary 2 (Compact metrizable spaces and separation). *If K is compact metrizable, then $L(K)$ has*

$$wsV^*_{(p,q)} \quad \text{and} \quad wsEV^*_{(p,q)}$$

for every $1 \leq p \leq q \leq \infty$. In particular, $L([0, 1])$ has all of these weak variants, but fails

$$V^*_{(p,q)}, \quad EV^*_{(p,q)}, \quad sV^*_{(p,q)}, \quad sEV^*_{(p,q)}$$

for every $1 \leq p \leq q \leq \infty$.

Proof. Since K is compact metrizable, $C(K)$ is separable. For each countable $S \subseteq K$, the restriction space

$$\overline{\{f|_S : f \in C(K)\}}^{\|\cdot\|_\infty}$$

is the closure of a continuous image of $C(K)$, and is therefore separable. The weak variants follow from Corollary 1.

For $K = [0, 1]$, the space K is infinite and functionally bounded. Theorem 1 therefore gives the failure of the four stronger variants listed above. $\square$

Limitations and Novelty Check

Theorem 1 characterizes the compact and sequentially compact V^* -type variants V^* , EV^* , sV^* , and sEV^* . Corollary 2 shows that the weaker wsV^* and $wsEV^*$ variants have a different characterization: finite functional boundedness is sufficient but not necessary.

A further push on the missing converse for the weak variants did not produce a justified full characterization. The operator characterization of wsV^* in [2] reduces the missing direction to constructing, from a bad countable functionally bounded support, a (q, p) -convergent adjoint $C(X) \rightarrow \ell_\infty$ whose associated sequence in $L(X)$ has no weakly Cauchy subsequence. The obvious candidates fail in opposite ways: point masses and block averages over $\beta\mathbb{N}$ have no weakly Cauchy subsequences, but coordinate or block indicator functions are weakly 1-summable and violate the V^* -test; summable-weight sign families satisfy the V^* estimates, but diagonal selection and dominated convergence give weakly Cauchy subsequences.

An earlier packet proved only the sufficient direction for ordinary V^* -sets, and a later packet recorded only Theorem 2. The present consolidated packet supersedes both.

Novelty checks were bounded but targeted. I checked the local run indexes for arXiv:2406.07178 and the free-lcs V^* keywords, the source paper [1], Gabrielyan’s Pełczyński-set preprint [2], and Gabrielyan’s limited-set preprint [3]. I also searched the web for the exact Problem 3.18 wording and close variants such as “ V^* free locally convex space $L(X)$ ” and “Problem 3.18 Gelfand–Phillips type properties”. No later paper explicitly answering this free-lcs V^* question was found in that bounded search.

Human Review Recommendation

There are two key points to check. First, in Lemma 1, verify that the weighted partial-sum operator $H : \ell_\infty \rightarrow \ell_\infty$ has the asserted convergence property on weakly p -summable sequences. Once that is confirmed, the finite-functional-boundedness characterization follows from standard free-lcs bounded-set arguments.

Second, in Theorem 2, verify that passing from a bounded sequence in $L(X)$ to the weak-star compact metrizable dual ball of E_S is legitimate and that weak-star convergence in E'_S gives weak Cauchyness in $L(X)$. This is the entire mechanism behind the positive weak-variant criterion and the $X = [0, 1]$ separation.

References

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